Type MS · The final 20-mark program

Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 1 of 7 · find part whole in this mark scheme extract

6	Award **one** mark for each of:			
	A1	Any variable for tracking best breed	Initial values could be the first breed in the table	
	A2	A loop to all items in an array	The same index can be used across all the individual arrays, regardless of loop type	
	A3	Daily volume calculation is volume * count	Disregard accuracy of indexing	
	A4	Rows of arrays are displayed	Disregard presence/lack of calculated daily volume	
	A5	Total volume calculated correctly by any method		
	A6	Relational operators used to compare rating and daily volume		
	A7	Boolean operator/nested selection used to combine tests		
	A8	Add calculated daily volume to the new data structure		
	A9	Description of data fields displayed (no alignment required)		
	A10	Recommended breed identified correctly	by any method except hard-coded	
	A11	Outputted information is fit for purpose and suitable for the audience		**11**

Type MS · The final 20-mark program

Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 2 of 7 · find part whole in this mark scheme extract

Award up to a maximum of nine marks using the levels-based mark scheme below.				Mark
Band 0	**Band 1 (1-3 marks)**	**Band 2 (4-6 marks)**	**Band 3 (7-9 marks)**	
No rewardable content	Little attempt to decompose the problem into component parts	Some attempt to decompose the problem into component parts	The problem has been decomposed into component parts	
	Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
	Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
	Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
	Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
	The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	**(9)**

Type MS · The final 20-mark program

Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 3 of 7 · find part whole in this mark scheme extract

C#

```
// Q06FINISHED

string[] tbl_breed = { "Red Chittagong", "Sussex", "Dexter",
                       "Abondance", "Sahiwal", "Vorderwald",
                       "Ayrshire", "Jersey", "Randall",
                       "Alderney", "Carora", "Gloucester"};
int[] tbl_rating = { 1, 2, 3, 2, 3, 1, 2, 1, 2, 1, 3, 2 };
int[] tbl_count = { 6, 3, 8, 7, 6, 4, 3, 7, 3, 3, 4, 7 };
double[] tbl_volume = {7.5, 5.7, 11.4, 11.4,
                       22.0, 15.2, 21.0, 18.3,
                       19.0, 9.0, 23.1, 16.0 };
double[] tbl_dailyVolume = {0.0, 0.0, 0,0, 0,0,
                            0.0, 0.0, 0.0, 0.0,
                            0.0, 0.0, 0.0, 0.0};

// ---------------------------------------------
// Write your code below this line

const String SPACE = " ";           // Just for reading

// Variables for indexing and totals
double totalVolume = 0.0;
double todayVolume = 0.0;

// Variables for outputting
string outString = "";

// Initialise the maximum values to the first instance
int maxIndex = 0;

// Display a message to describe the data fields
Console.WriteLine ("Fields are: Breed, Rating, Volume (cow), Count, Volume (day)");
```

```
// Process every breed of cow
for (int i = 0; i < tbl_breed.Length; i++)
{
    // Calculate volume per day
    todayVolume = tbl_volume[i] * tbl_count[i];

    // Add today's volume to the daily volume table
    tbl_dailyVolume[i] = todayVolume;

    // Display the row of information for this breed
    outString = tbl_breed[i] + SPACE +
                tbl_rating[i].ToString() + SPACE +
                tbl_volume[i].ToString() + SPACE +
                tbl_count[i].ToString() + SPACE +
                tbl_dailyVolume[i].ToString();
    Console.WriteLine(outString);

    // Keep a running total of milk production
    totalVolume = totalVolume + todayVolume;

    // Test for a best breed
    if ((tbl_rating[i] <= tbl_rating[maxIndex]) &
        (tbl_volume[i] >= tbl_volume[maxIndex]))
    {
        maxIndex = i;
    }
}

// Display the total volume of milk in a day
outString = "Total: " + totalVolume.ToString() + " litres";
Console.WriteLine(outString);

// Display the recommended breed
outString = "Recommended breed: " + tbl_breed[maxIndex] +
            " rating: " + tbl_rating[maxIndex].ToString() +
            " volume: " + tbl_volume[maxIndex];
Console.WriteLine(outString);
```

Type MS · The final 20-mark program

Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 4 of 7 · find part whole in this mark scheme extract

Java

```java
// Q06FINISHED

public class Q06FINISHED
{
    static final String SPACE = " ";            // Just for reading

    public static void main(String[] args)
    {
        String[] tbl_breed = { "Red Chittagong", "Sussex", "Dexter", "Abondance",
                               "Sahiwal", "Vorderwald", "Ayrshire", "Jersey",
                               "Randall", "Alderney", "Carora", "Gloucester"};
        int[] tbl_rating = { 1, 2, 3, 2, 3, 1, 2, 1, 2, 1, 3, 2 };
        int[] tbl_count = { 6, 3, 8, 7, 6, 4, 3, 7, 3, 3, 4, 7 };
        double[] tbl_volume = { 7.5, 5.7, 11.4, 11.4,
                                22.0, 15.2, 21.0, 18.3,
                                19.0, 9.0, 23.1, 16.0 };
        double [] tbl_dailyVolume = {0.0, 0.0, 0,0, 0,0,
                                     0.0, 0.0, 0.0, 0.0,
                                     0.0, 0.0, 0.0, 0.0};

        // ------------------------------------------------
        // Write your code below this line

        // Variables for indexing and totals
        double totalVolume = 0.0;
        double todayVolume = 0.0;

        // Variables for outputting
        String outString = "";

        // Initialise the maximum values to the first instance
        int maxIndex = 0;

       // Display the titles in a table
        System.out.println ("Fields are: Breed, Rating, Volume (cow), Count, Volume (day)");

```

Type MS · The final 20-mark program

Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 5 of 7 · find part whole in this mark scheme extract

```
37          // Process every breed of cow
38          for (int i = 0; i < tbl_breed.length; i++)
39          {
40              // Calculate volume per day
41              todayVolume = tbl_volume[i] * tbl_count[i];
42
43              // Add today's volume to the daily volume table
44              tbl_dailyVolume[i] = todayVolume;
45
46              // Display the row of information for this breed
47              outString = tbl_breed[i] + SPACE +
48                          Integer.toString(tbl_rating[i]) + SPACE +
49                          String.valueOf(tbl_volume[i]) + SPACE +
50                          Integer.toString(tbl_count[i]) + SPACE +
51                          String.valueOf(tbl_dailyVolume[i]);
52              System.out.println (outString);
53
54              // Keep a running total of milk production
55              totalVolume = totalVolume + todayVolume;
56
57              // Test for a best breed
58              if ((tbl_rating[i] <= tbl_rating[maxIndex]) &
59                      (tbl_volume[i] >= tbl_volume[maxIndex]))
60              {
61                  maxIndex = i;
62              }
63          }
64
65          // Display the total volume of milk in a day
66          outString = "Total: " + Double.toString(totalVolume) + " litres";
67          System.out.println (outString);
68
69          // Display the recommended breed
70          outString = "Recommended breed: " + tbl_breed[maxIndex] +
71                      " rating: " +
72                      Integer.toString(tbl_rating[maxIndex]) +
73                      " volume: " +
74                      Double.toString(tbl_volume[maxIndex]);
75          System.out.println (outString);
76      }
77  }
```

Type MS · The final 20-mark program

Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 6 of 7 · find part whole in this mark scheme extract

Python

```
# Q06FINISHED

tbl_breed = ["Red Chittagong", "Sussex", "Dexter", "Abondance",
             "Sahiwal", "Vorderwald", "Ayrshire", "Jersey",
             "Randall", "Alderney", "Carora", "Gloucester"]
tbl_rating = [1, 2, 3, 2, 3, 1, 2, 1, 2, 1, 3, 2]
tbl_count = [6, 3, 8, 7, 6, 4, 3, 7, 3, 3, 4, 7]
tbl_volume = [7.5, 5.7, 11.4, 11.4,
              22.0, 15.2, 21.0, 18.3,
              19.0, 9.0, 23.1, 16.0]
tbl_dailyVolume = []

# ---------------------------------------------
# Write your code below this line

# Variables for indexing and totals
index = 0
totalVolume = 0.0

# Initialise the maximum values to the first instance
maxIndex = 0

# Display a message to describe the data fields
print ("Fields are: Breed, Rating, Volume (cow), Count, Volume (day)")

# Process every breed of cow
for index in range (len (tbl_breed)):
    # Calculate volume per day
    todayVolume =  tbl_volume[index] * tbl_count[index]

    # Add today's volume to the daily volume table
    tbl_dailyVolume.append (todayVolume)

    # Display the row of information for this breed
    print (tbl_breed[index],tbl_rating[index],
           tbl_volume[index], tbl_count[index], tbl_dailyVolume[index])

    # Keep a running total of milk production
    totalVolume = totalVolume + todayVolume

    # Test for a best breed
    if ((tbl_rating[index] <= tbl_rating[maxIndex]) and
                (tbl_volume[index] >= tbl_volume[maxIndex])):
        maxIndex = index

# Display the total volume of milk in a day
layout = "Total: {} litres"
print (layout.format (totalVolume))

# Display the recommended breed
print ("Recommended breed: " + tbl_breed[maxIndex],
       "rating: " + str(tbl_rating[maxIndex]),
       "volume " + str(tbl_volume[maxIndex]))
```

Type MS · The final 20-mark program

Answer 1 of 8 · Question 6 part whole · 20 marks

2023 JUN · Paper 02 · extract 7 of 7 · find part whole in this mark scheme extract

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Type MS · The final 20-mark program

Answer 2 of 8 · Question 5 part whole · 20 marks

2019 JUN · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

Q5	Designated marks: 1. Initialise variables for at least two of total number of books, count of pupils and average 2. Initialise variables for gold, silver and bronze / sort array in order of books read 3. Create loop for identifying (and printing) those borrowing fewer than 10 books 4. Print out appropriate details (must include pupil ID) 5. Print out total of books borrowed 6. Print out average number of books borrowed 7. Identification of gold medal winner 8. Identification of silver medal winner 9. Identification of bronze medal winner 10. Details of at least one of gold, silver and bronze medal winner printed (minimum last name or first name) 11. Details of all three medal winners (first name and last name)	Award marks independently Award marks for reasonable attempt to meet requirement.	(11)

Type MS · The final 20-mark program

Answer 2 of 8 · Question 5 part whole · 20 marks

2019 JUN · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Question 5 Overview marks:				
Mark Band 0	**Mark Band 1**	**Mark Band 2**	**Mark Band 3**	**Marks**
0	1 – 3	4 – 6	7 – 9	
No rewardable content	Little attempt to decompose the problem into component parts Some parts of the logic are clear and appropriate to the problem Some appropriate use and manipulation of data types, variables, data structures and program constructs Parts of the code are clear and readable Finished program will not be flexible enough with other data sets or input The program meets some of the given requirements	Some attempt to decompose the problem into component parts Most parts of the logic are clear and mostly appropriate to the problem The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate Code is mostly clear and readable Finished program will function with some but not all other data sets or input The program meets most of the given requirements	The problem has been decomposed into component parts The logic is clear and appropriate to the problem The use and manipulation of data types, variables and data structures and program constructs is appropriate Code is clear and readable Finished program could be used with other data sets or input The program fully meets the given requirements	(9)

Type MS · The final 20-mark program

Answer 2 of 8 · Question 5 part whole · 20 marks

2019 JUN · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

C# Example code

Type MS · The final 20-mark program

Answer 3 of 8 · Question 6 part whole · 20 marks

2020 NOV · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

6	A1	2.2.1	Initialise variable for total sales by member of staff (1)	Sales for each member of staff George Taylor 28466 Fehlix Chayne 35250 Sumren Bergen 30186 Samira Beckle 37430 Nellie Bogart 20448 Cheryl Grouth 34383 Danuta Graunt 26421 Jaiden Deckle 21507 Jimran Caliks 32415 Deynar Derran 28590 ========== Total staff sales: 295096 Highest sales by: Samira Beckle with 37430 Second highest sales by: Fehlix Chayne with 35250 >>>	(11)
	A2	2.5.1	Add succeeding months' sales to total sales (1)		
	A3	2.1.6	Display total sales by member of staff (1)		
	A4	2.2.2	Loop repeats for each member of staff (1)		
	A5	2.2.1	Running team total sales initialised (1)		
	A6	2.2.2	Add succeeding individual staff sales to team total (1)		
	A7	2.1.6	Display final running total (1)		
	A8	2.1.6	Staff member with highest individual sales identified (1)		
	A9	2.1.6	Staff member with second highest individual sales identified (1)		
	A10	2.1.6	Information about staff with highest sales displayed (1)		
	A11	2.1.6	Information about staff with second highest sales displayed (1)		

14 of 16

Type MS · The final 20-mark program

Answer 3 of 8 · Question 6 part whole · 20 marks

2020 NOV · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Band 1 (1-3 marks)	Band 2 (4-6 marks)	Band 3 (7-9 marks)	Mark
Little attempt to decompose the problem into component parts	Some attempt to decompose the problem into component parts	The problem has been decomposed into component parts	
Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	(9)

15 of 16

Type MS · The final 20-mark program

Answer 3 of 8 · Question 6 part whole · 20 marks

2020 NOV · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Code example

Python

```
allStaffTotal = 0
highestSales = 0
secondSales = 0
high = 0
second = 0

print ("Sales for each member of staff","\n\n")
for staff in staffSales:
    staffTotal = 0
    sales = 3
    while (sales < 9):
        staffTotal = staffTotal + staff[sales]
        sales = sales + 1
    print (staff[1],staff[2],staffTotal)
    if (staffTotal > secondSales):
        if (staffTotal > highestSales):
            highestSales = staffTotal
            high = staff
        else:
            secondSales = staffTotal
            second = staff
    allStaffTotal = allStaffTotal + staffTotal
print ("==========\n\nTotal sales for all staff: ", allStaffTotal)

print ("\n\nHighest sales by: ",high[1],high[2], " with ", highestSales)
print ("\n\nSecond highest sales by: ",second[1],second[2], " with ", secondSales)
```

16 of 16

Type MS · The final 20-mark program

Answer 4 of 8 · Question 5 part whole · 20 marks

2021 JUN · Paper 02 · extract 1 of 5 · find part whole in this mark scheme extract

Question		Answer	Additional Guidance	Mark
5		**addPlayerName()**		
	A1	Suitable prompt for player name and assigned to suitable variable		
		guessCapital()		
	A2	Ensure question can only be used once	Do not award if more than one question variable e.g. question1, question2 etc.	
	A3	Question includes suitable message and country name		
	A4	Check made to see if guess is correct		
	A5	If guess correct score incremented		
	A6	If guess is incorrect suitable message displayed		
	A7	If guess incorrect capital concatenated with message		
	A8	Repeated for a minimum of five questions	Do not award if questions are asked manually e.g. question1, question2, repeated code for each question etc. Do not award if 5 unique questions are not asked while the program is running	
		Main Program		
	A9	Player name or score displayed	Do not award if the return value from at least one function is not used	
	A10	At least one menuChoice calls correct subprogram		
	A11	Main program calls the two sub-programs correctly		(11)

Type MS · The final 20-mark program

Answer 4 of 8 · Question 5 part whole · 20 marks

2021 JUN · Paper 02 · extract 2 of 5 · find part whole in this mark scheme extract

Band 1 (1-3 marks)	Band 2 (4-6 marks)	Band 3 (7-9 marks)	Mark
Little attempt to decompose into component parts	Some attempt to decompose into component parts	The problem has been decomposed into component parts	
Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	(9)

Type MS · The final 20-mark program

Answer 4 of 8 · Question 5 part whole · 20 marks

2021 JUN · Paper 02 · extract 3 of 5 · find part whole in this mark scheme extract

Code examples

Python

Add player name function

```
    player = ""
    while(player == ""):
        player = input("Enter your player name ")
    return player
#-----------------------------------------------------
```

Main program

```
menuChoice = 0
score = 0
playerName = ""

while menuChoice != 3:
    displayMenu()
    menuChoice = getMenuChoice()

    # Add your code to:
    #   call the relevant subprogram if the menu choice is 1 or 2
    #   display the player name and the score if the menu choice is 3
    if menuChoice == 1:
        playerName = addPlayerName()
    elif menuChoice == 2:
        score = guessCapital()
    else:
        print("Well done "+ playerName + ". The score is " + str(score))
```

Type MS · The final 20-mark program

Answer 4 of 8 · Question 5 part whole · 20 marks

2021 JUN · Paper 02 · extract 4 of 5 · find part whole in this mark scheme extract

Guess capital city function

```
# Add your code here
questionCount = 1
questionScore = 0

# Ask 5 questions
while questionCount <= 5:

    questionChoice = -1
    questionNumbers = ""

    # Build a string containing the question numbers available
    for question in questions:
        if question != 0:
         questionNumbers += str(question) + " "

    # Ensure valid question number is chosen
    while str(questionChoice) not in questionNumbers:
        questionChoice = int(input("Pick a number from " + questionNumbers))

    # Get the country and its capital
    country = countries[questionChoice - 1]
    capital = capitals[questionChoice - 1]

    # Display the country and get the guess
    guess = input("What is the capital of " + country + "? ").lower()

    # If the guess is correct display message and increment score
    if guess == capital.lower():
        print("Well done, you guessed correctly")
        questionScore += 1
    else:
        # Otherwise display the country name and correct capital
        print("You did not guess correctly. The capital of " + country + " is " + capital)

    # Increment the number of questions asked
    questionCount = questionCount + 1

    # Set the question number to 0 so that it cannot be guessed again
    questions[questionChoice - 1] = 0;
    questions[questionChoice - 1] = 0;

# return the score to the main menu
return questionScore
```

Type MS · The final 20-mark program

Answer 4 of 8 · Question 5 part whole · 20 marks

2021 JUN · Paper 02 · extract 5 of 5 · find part whole in this mark scheme extract

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Type MS · The final 20-mark program

Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 1 of 8 · find part whole in this mark scheme extract

5	A1	At least three variables initialised appropriately		(1)
	A2	Word input		(1)
	A3	Program repeats until 1 is input		(1)
	A4	Loop through each word in the array		(1)
		Words that begin with the same letter		
	A5	Check to see if the first letter of the input word matches the first letter of at least one word in the array		(1)
	A6	The number of words that begin with the same letter displayed		(1)
		Words that contain the input word		
	A7	Check to see if the word contains the input word		(1)
	A8	Number of letters in the word calculated		(1)
	A9	At least one word from the array that contains the input word displayed		(1)
	A10	The number of words that contain the input word displayed		(1)
	A11	Identification of the longest word or the shortest word		(1)

Band 1 (1-3 marks)	Band 2 (4-6 marks)	Band 3 (7-9 marks)	Mark
Little attempt to decompose into component parts	Some attempt to decompose into component parts	The problem has been decomposed into component parts	
Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	(9)

Type MS · The final 20-mark program

Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 2 of 8 · find part whole in this mark scheme extract

Code examples

C#

```
// Add your code here
string inputWord = "";

while (inputWord != "1")
{
    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    Console.WriteLine("Enter a word or 1 to exit ");
    inputWord = Console.ReadLine();

    if (inputWord != "1")
    {
        int count = 0;

        foreach (String word in wordArray)
        {
            if(word[0] == inputWord[0])
            {
                Console.WriteLine(word);
                count ++;
            }
        } // End of Loop

        Console.WriteLine(count + " word(s) beginning with "+ inputWord[0]);
        Console.WriteLine("----------------------------------------------------------------------");

        count = 0;

        foreach (String word in wordArray)
        {
            if(word.Contains(inputWord))
            {
                count ++;
                Console.WriteLine(word);

                int length = word.Length;

                if (longest < length)
```

Type MS · The final 20-mark program

Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 3 of 8 · find part whole in this mark scheme extract

```
                    {
                        longest = length;
                        longestWord = word;
                    }

                    if (shortest > length)
                    {
                        shortest = length;
                        shortestWord = word;
                    }
                }// end of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word

            if (count > 0)
            {
                Console.WriteLine(count + " word(s) with "+ inputWord + " in them");
                Console.WriteLine("The longest word has " + longest + " characters");
                Console.WriteLine("The shortest word has " + shortest + " characters");
                Console.WriteLine("The longest word is " + longestWord + " characters");
                Console.WriteLine("The shortest word is " + shortestWord + " characters");
                Console.WriteLine("--------------------------------------------------------------------------------");
            }
            else
            {
                Console.WriteLine("There were 0 words that had all the letters from " + inputWord + " in them");
                Console.WriteLine("--------------------------------------------------------------------------------");
            } // End of printing words
        } // End of check
    } // End of while loop
```

Type MS · The final 20-mark program

Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 4 of 8 · find part whole in this mark scheme extract

Java

```
// Add your code here
Scanner input = new Scanner(System.in);
String inputWord = "";

// While there is a word run the program
while (!inputWord.equals("1")) {

    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    System.out.println("Enter a word or 1 to exit ");
    System.out.println("--------------------------------------------------------------------------------");
    inputWord = input.nextLine();
    if (!inputWord.equals("1")){
        // Find words that begin with the same letter as the input word
        int count = 0;

        // Get each word in the array
        for (String word: wordArray){

            // If the first character of the word is the same as the first character of
            // the input word then print the word
            if (word.charAt(0) == inputWord.charAt(0)){
                System.out.println(word);
                count ++;
            }
        } // End of for loop

        // Print the total number of words that begin with the same letter
        System.out.println(count + " word(s) beginning with "+ inputWord.charAt(0));
        System.out.println("--------------------------------------------------------------------------------");

        // Find words that contain the input word
        count = 0;

        // Get each word in the array
        for (String word: wordArray){
            // Check to see if the word contains the inputWord
            if (word.contains(inputWord)){
                count ++;
                System.out.println(word);

                // Set the length of the word
                int length = word.length();
```

Type MS · The final 20-mark program

Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 5 of 8 · find part whole in this mark scheme extract



```
                    // If the length is longer than the current longest replace it and the longestWord
                    if (longest < length) {
                        longest = length;
                        longestWord = word;
                    }

                    // If the length is shorter than the shortest replace it and the shortestWord
                    if (shortest > length){
                        shortest = length;
                        shortestWord = word;
                    }
                } // End of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word
            if( count > 0){
                System.out.println(count + " word(s) with "+ inputWord + " in them");
                System.out.println("The longest word has " + longest + " characters");
                System.out.println("The shortest word has " + shortest + " characters");
                System.out.println("The longest word is " + longestWord + " characters");
                System.out.println("The shortest word is " + shortestWord + " characters");
                System.out.println("--------------------------------------------------------------------------------");
            } else {
                System.out.println("There were 0 words that had all the letters from " + inputWord + " in them");
                System.out.println("--------------------------------------------------------------------------------");
            } // End of printing words
        }
    } // End of while loop
```

Type MS · The final 20-mark program

Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 6 of 8 · find part whole in this mark scheme extract

Python

```
# While there is a word run the program
while inputWord !="1":

    shortest = 5000
    longest = 0
    shortestWord = ""
    longestWord = ""

    inputWord = input("Enter a word or 1 to exit: ")
    print("------------------------------------------------------------------------")

    if inputWord !="1":
        # Find words that begin with the same letter as the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # If the first character of the word is the same as the first character of
            # the input word then print the word
            if word[0] == inputWord[0]:
                print(word)
                count = count + 1

        # Print the total number of words that begin with the same letter
        print("\n",count,"word(s) beginning with",inputWord[0])
        print("------------------------------------------------------------------------")

        # Find words that contain the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # Check to see if the word contains the inputWord
            if inputWord in word:
                count = count + 1
                print(word)
                # Set the length of the word
                length = len(word)

                # If the length is longer than the current longest replace it and the longestWord
                if longest < length:
                    longest = length
                    longestWord = word

                # If the length is shorter than the shortest replace it and the shortestWord
                if shortest > length:
                    shortest = length
                    shortestWord = word
```

Type MS · The final 20-mark program

Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 7 of 8 · find part whole in this mark scheme extract

```
59
60          # Print the number of words, the number of characters in the longest and
61          # shortest word, the longest and the shortest word
62          if count > 0:
63              print("\n",count,"word(s) with",inputWord,"in them")
64              print("The longest word has",longest,"letters")
65              print("The shortest word has",shortest,"letters")
66              print("The longest word is",longestWord)
67              print("The shortest word is",shortestWord)
68              print("-------------------------------------------------------------------------")
69          else:
70              print("There were 0 words with all the letters from",inputWord,"in them")
71          # ------------------------------------------------------------------------------------
72
73  # End the program when there are no more words
74  print("End of program")
```

Type MS · The final 20-mark program

Answer 5 of 8 · Question 5 part whole · 20 marks

2021 NOV · Paper 02 · extract 8 of 8 · find part whole in this mark scheme extract

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Answer 6 of 8 · Question 6 part whole · 20 marks

2022 JUN · Paper 02 · extract 1 of 4 · find part whole in this mark scheme extract

6	A1	At least two appropriate subprograms used		(1)
	A2	Repeat guess until the word has been guessed **OR** no attempts left		(1)
	A3	Generate and display the number of attempts left	Must see working correctly	(1)
	A4	Request input of the word		(1)
	A5	Validate the input of the word to ensure it is the same length as the random word		(1)
	A6	Check to see if the guess matches the random word		(1)
	A7	If the word has been guessed, display message including the random word and number of attempts		(1)
	A8	Keep track and display letters that are in the word	Must see working	(1)
	A9	Keep track and display letters that are not in the word	Must see working	(1)
	A10	Letters appear only once in each list	Must see working	(1)
	A11	Display lose message including the random word		(1)

Band 1 (1-3 marks)	Band 2 (4-6 marks)	Band 3 (7-9 marks)	Mark
Little attempt to decompose into component parts	Some attempt to decompose into component parts	The problem has been decomposed into component parts	
Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	(9)

Type MS · The final 20-mark program

Answer 6 of 8 · Question 6 part whole · 20 marks

2022 JUN · Paper 02 · extract 2 of 4 · find part whole in this mark scheme extract

Code examples

Python

Subprograms

```
# Add your subprograms here
def checkInput(pWordLength):
    # Validate input to make sure it is the same length as the secret word
    check = ""
    while len(check) != pWordLength:
        check = input("Enter your guess. The secret word has " +str(pWordLength)+ " letters: ")
    return(check)

def checkLettersInWord(pGuess, pWordToGuess, pLetters, type):
    for letter in pGuess: # Check each letter in the guess
        if type == 0: # The check to carry out is for correct letters
            # If the letter is in the secret word and not already in the correctletters, add it
            if letter in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
        else: # The check to carry out is for wrong letters
            # If the letter is not in the secret word and not already in the wrongLetters, add
            if letter not in pWordToGuess and letter not in pLetters:
                pLetters = pLetters + letter
    return pLetters

def display(pCorrectLetters, pWrongLetters):
    # Display the correct and wrong letters
    if len(pCorrectLetters) > 0:
        print("Letter(s) in the secret word:",pCorrectLetters)

    if len(pWrongLetters) > 0:
        print("Letter(s) not in the secret word:",pWrongLetters)
```

Type MS · The final 20-mark program

Answer 6 of 8 · Question 6 part whole · 20 marks

2022 JUN · Paper 02 · extract 3 of 4 · find part whole in this mark scheme extract

Main program

```
# Add your main program code here
wordLength = len(wordToGuess)
maxAttempts = wordLength + 3
numAttempts = 0
correctLetters = ""
wrongLetters = ""
guessed = False

# Loop until the no attempts are left or the secret word has been guessed
while numAttempts < maxAttempts and guessed == False:
    print("You have",maxAttempts - numAttempts,"attempts to guess the secret word.")
    numAttempts = numAttempts + 1

    guess = checkInput(wordLength)  # Validate the input

    if guess == wordToGuess: # Check to see if the secret word has been guessed
        guessed = True
        print("Well done. You guessed the secret word was",wordToGuess,"in",numAttempts,"attempts.")

    else: # If it hasn't been guessed check for correct and wrong letters
        correctLetters = checkLettersInWord(guess,wordToGuess, correctLetters, 0)
        wrongLetters = checkLettersInWord(guess,wordToGuess, wrongLetters, 1)
        display(correctLetters, wrongLetters)

# Display game over message if attempts have run out and the secret word has not been guessed
if numAttempts == maxAttempts and guessed == False:
    print("Game over. You did not guess that the secret word was",wordToGuess)
```

Type MS · The final 20-mark program

Answer 6 of 8 · Question 6 part whole · 20 marks

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Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 1 of 10 · find part whole in this mark scheme extract

6		Award **one** mark for each of:		
	A1	Put new group of words (newspaper, magazine) into last blank position in array (1)	Any method Do not award hard-coded changes to table Note: *if* the solution will not work for any length of array. This will impact LBMS functionality.	
	A2	Loop for processing every group of words in the array (1)	Allow hard-coded length	
	A3	Loop controlled by the length of the array (1)	Method/function to determine the number of groups in the array	
	A4	Mechanism for tracking group number to appear in output (1)	Any method	
	A5	Output includes row number and original group of words (1)	Ignore 'punctuation' Requires both a line number and each word pair	
	A6	Mechanism for finding longest word in each group (1)	Method/function to determine the length of each string	
	A7	Mechanism for determining if the words in the group are out of order (1)	Any method	
	A8	Mechanism for putting the words in the group into alphabetical order (1)	Any method	
	A9	Output includes longest word in each group and any group found to be out of order is displayed in order (1)		
	A10	Evidence of two-dimensional indexing (1)	Any method	**11**

Type MS · The final 20-mark program

Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 2 of 10 · find part whole in this mark scheme extract

	A11	At least one instance of an informative comment (1)		

Type MS · The final 20-mark program

Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 3 of 10 · find part whole in this mark scheme extract

Award up to a maximum of nine marks using the levels-based mark scheme below.				Mark
Band 0	**Band 1 (1-3 marks)**	**Band 2 (4-6 marks)**	**Band 3 (7-9 marks)**	
No rewardable content	Little attempt to decompose the problem into component parts	Some attempt to decompose the problem into component parts	The problem has been decomposed into component parts	
	Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
	Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
	Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
	Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
	The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	**(9)**

Type MS · The final 20-mark program

Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 4 of 10 · find part whole in this mark scheme extract

C#

```
// Q06FINISHED

String[,] tblWords = {{"apple", "banana"},
                      {"wrist", "leg"},
                      {"blue", "yellow"},
                      {"speaker", "keyboard"},
                      {"lavender", "tulip"},
                      {"pencil", "chalk"},
                      {"apartment", "house"},
                      {"bottom", "top"},
                      {"snow", "fog"},
                      {"beach", "mountain"},
                      {"", ""}};

String word1 = "newspaper";
String word2 = "book";

// -----------------------------------------
// Write your code below this line
int numRows = tblWords.GetLength(0);
String outString = "";              // For output
int indexMaxLength = 0;             // For checking lengths
String temp = "";                    // For swapping

// Insert a new group of words into the blank row
tblWords[numRows - 1, 0] = word1;
tblWords[numRows - 1, 1] = word2;

```

Type MS · The final 20-mark program

Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 5 of 10 · find part whole in this mark scheme extract

```
// Loop for each individual group of words in the table
for (int i = 0; i < numRows; i++)
{
    // Pick up an entire row
    String[] words;
    words = new String[2];
    words[0] = tblWords[i, 0];
    words[1] = tblWords[i, 1];

    // Display the row number and the original content
    outString = Convert.ToString (1 + i);
    outString = outString + " " + words[0] + " " + words[1];
    Console.WriteLine (outString);

    // Find the length of the longest word and display the word
    indexMaxLength = 0;             // Longest is first word
    if (words[1].Length > words[indexMaxLength].Length)
    {
        indexMaxLength = 1;
    }
    // Display indented
    Console.WriteLine ("\t" + words[indexMaxLength]);

    // Check if the words in this group are in order
    if (words[0][0] > words[1][0])
    {
        // This group is out of order so needs ordering
        temp = words[0];
        words[0] = words[1];
        words[1] = temp;

        Console.WriteLine ("\t" + words[0] + " " + words[1]);
    }
}
```

Type MS · The final 20-mark program

Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 6 of 10 · find part whole in this mark scheme extract

Java

```java
// Q06FINISHED

public class Q06FINISHED
{
    public static void main(String[] args)
    {
        String[][] tblWords = {{"apple", "banana"},
                               {"wrist", "leg"},
                               {"blue", "yellow"},
                               {"speaker", "keyboard"},
                               {"lavender", "tulip"},
                               {"pencil", "chalk"},
                               {"apartment", "house"},
                               {"bottom", "top"},
                               {"snow", "fog"},
                               {"beach", "mountain"},
                               {"", ""}};

        String word1 = "newspaper";
        String word2 = "book";

        // --------------------------------------------
        // Write your code below this line
        int numRows = tblWords.length;
        String outString = "";              // For output
        int indexMaxLength = 0;             // For checking lengths
        String temp = "";                    // For swapping

        // Insert a new group of words into the blank row
        tblWords[numRows - 1][0] = word1;
        tblWords[numRows - 1][1] = word2;

```

Type MS · The final 20-mark program

Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 7 of 10 · find part whole in this mark scheme extract

```
        // Loop for each individual group of words in the table
        for (int i = 0; i < numRows; i++)
        {
            // Pick up an entire row
            String[] words = tblWords[i];

            // Display the row number and the original content
            outString = String.valueOf(1 + i) + " " +  words[0] + " " + words[1];
            System.out.println (outString);

            // Find the length of the longest word and display the word
            indexMaxLength = 0;             // Longest is first word
            if (words[1].length() > words[indexMaxLength].length())
            {
                indexMaxLength = 1;
            }

            // Display indented
            System.out.println ("\t" + words[indexMaxLength]);

            // Check if the words in this group are in order
            if (words[0].charAt (0) > words[1].charAt(0))
            {
                // This group is out of order so needs ordering
                temp = words[0];
                words[0] = words[1];
                words[1] = temp;

                System.out.println ("\t" + words[0] + " " + words[1]);

            }
        }
    }
}
```

Type MS · The final 20-mark program

Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 8 of 10 · find part whole in this mark scheme extract

Python

```
# Q06FINISHED

tblWords = [["apple", "banana"],
            ["wrist", "leg"],
            ["blue", "yellow"],
            ["speaker", "keyboard"],
            ["lavender", "tulip"],
            ["pencil", "chalk"],
            ["apartment", "house"],
            ["bottom", "top"],
            ["snow", "fog"],
            ["beach", "mountain"],
            ["", ""]]

word1 = "newspaper"
word2 = "book"

# ------------------------------------------------
# Write your code below this line
row = 1                 # Row number
numRows = len (tblWords)

# Insert a new group of words into the blank row
tblWords[numRows - 1][0] = word1
tblWords[numRows - 1][1] = word2

# Loop for each individual group of words in the table
for words in tblWords:
    # Display the row number and the original content
    print (str (row) + " " + words[0] + " " + words[1])

    # Find the length of the longest word and display the word
    indexMaxLength = 0          # Longest is first word
    if (len (words[1]) > len (words[indexMaxLength])):
        indexMaxLength = 1

    # Display indented
    print (" "*5 + words[indexMaxLength])

    # Check if the words in this group are in order
    if (words[0] > words[1]):
        # This group is out of order so needs sorting
        temp = words[0]
        words[0] = words[1]
        words[1] = temp
```

Type MS · The final 20-mark program

Answer 7 of 8 · Question 6 part whole · 20 marks

2024 JUN · Paper 02 · extract 9 of 10 · find part whole in this mark scheme extract

Output

```
1 apple banana
    banana
2 wrist leg
    wrist
    leg wrist
3 blue yellow
    yellow
4 speaker keyboard
    keyboard
    keyboard speaker
5 lavender tulip
    lavender
6 pencil chalk
    pencil
    chalk pencil
7 apartment house
    apartment
8 bottom top
    bottom
9 snow fog
    snow
    fog snow
10 beach mountain
```

Type MS · The final 20-mark program

Answer 7 of 8 · Question 6 part whole · 20 marks

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Answer 8 of 8 · Question 6 part whole · 20 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part whole in this mark scheme extract

6		Award **one** mark for each of:		
	A1	Both City and Temp taken as input (1)		
	A2	Both correct conversion calculation coded (1)	Does not have to be used	
	A3	Any relevant subprogram has been declared (1)	At least one **relevant** subprogram declared. `def name(param):` or `def name():` **and** one line of code indented	
	A4	Temperature in correct parallel position is updated when existing city is input (1)		
	A5	Loop for accessing all items in either array (1)	**For each** acceptable for this mark	
	A6	Table includes header and dividing line (1)	`print("city Celsius fahrenheit")` `print("-------------")`	
	A7	Mechanism for getting temp from string and completing conversion (1)	Include int( ) and passed to formula for at least one temperature value	
	A8	Mechanism for getting scale C/F (1)		
	A9	All rows output to table with corresponding city and (at least one) temperature values (1)	(same index value used) do not accept hard coded	
	A10	Table (for cities) all correct (1)	Ignore header (A6) and average (A11) Must have city, C value and F value (all accurate) Ignore float values if calculation correct	
	A11	Overall average correctly calculated and displayed in both scales (1)	Average may change depending on whether whole number cast or rounded – award accurate formula, not result.	**11**

20 of 23

Type MS · The final 20-mark program

Answer 8 of 8 · Question 6 part whole · 20 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part whole in this mark scheme extract

Paper 2 – Application of Computational Thinking

LBMS (AOs: 3xAO3.2a, 3xAO3.2b, 3xAO3.2c) (Tgt: 3, 3, 3); Spec: 1.1.6(2), 1.2.1(2), 1.2.2(2), 2.1.1, 2.1.2, 2.1.5
PBMS: 2.1.1, 2.3.2(2), 2.3.3(2), 2.4.1(2), 2.5.1(2), 2.6.1(2)

Award up to a maximum of nine marks using the levels-based mark scheme below.				Mark
Band 0	**Band 1 (1-3 marks)**	**Band 2 (4-6 marks)**	**Band 3 (7-9 marks)**	
No rewardable content	Little attempt to decompose the problem into component parts	Some attempt to decompose the problem into component parts	The problem has been decomposed into component parts	
	Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
	Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
	Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
	Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
	The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	**(9)**

21 of 23

Type MS · The final 20-mark program

Answer 8 of 8 · Question 6 part whole · 20 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part whole in this mark scheme extract

Paper 2 – Application of Computational Thinking

```
# Q06

# city and temperature arrays
Tbl_city = ["Reggane", "Cairo", "New Delhi", "Muscat", "Athens",
            "Barcelona", "Havana", "Phoenix", "Brisbane", "Darwin"]
Tbl_temps = ["F103", "F82", "C34", "F95", "C29", "F79", "F84",
             "C35", "F77", "F81"]

# ----------------------------------------------
# write subprogam(s) below this line
def CtoF(pTemp):
    fhTemp = pTemp * 1.8 + 32
    fhTemp = int(round(fhTemp,0))
    return fhTemp

def FtoC(pTemp):
    ctTemp = (pTemp - 32) / 1.8
    ctTemp = int(round(ctTemp,0))
    return ctTemp

# ----------------------------------------------
# Write your code below this line
totalF = 0
totalC = 0
cTemp = 0
fTemp = 0

cityCount = len(Tbl_city)

# update city temp
thisCity = input("Please enter City: ")
for x in range(cityCount):
    if thisCity == Tbl_city[x]:
        thisTemp = input("Please enter temp in correct format: ")
        Tbl_temps[x] = thisTemp
else:
    print("City does not exist")

# display table header
print("City          Celsius          Fahrenheit")
print("-----------------------------------------------------")

# take each city and temp
for i in range(cityCount):

    # get each scale and temp
    nextTemp = Tbl_temps[i]
    scale = nextTemp[0]
    theTemp = int(nextTemp[1:])

    # convert between temp scales
    if scale == "C":
        cTemp = theTemp
        totalC = totalC + cTemp
        fTemp = CtoF(theTemp)
        totalF = totalF + fTemp
    else:
        fTemp = theTemp
        totalF = totalF + fTemp
        cTemp = FtoC(theTemp)
        totalC = totalC + cTemp

    # display each row
    print(Tbl_city[i], "          ", cTemp, "          ", fTemp)

# calculate averages
averageF = int(round(totalF / cityCount,0))
averageC = int(round(totalC / cityCount,0))

# display table footer
print("-"*40)
print("Overall Average      ", averageC, "          ", averageF)
```

22 of 23

Type MS · The final 20-mark program

Answer 8 of 8 · Question 6 part whole · 20 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part whole in this mark scheme extract

Paper 2 – Application of Computational Thinking

```
# Q06

# city and temperature arrays
Tbl_city = ["Reggane", "Cairo", "New Delhi", "Muscat", "Athens",
            "Barcelona", "Havana", "Phoenix", "Brisbane", "Darwin"]
Tbl_temps = ["F103", "F82", "C34", "F95", "C29", "F79", "F84",
             "C35", "F77", "F81"]

# -----------------------------------------------------
# write subprogam(s) below this line
def CtoF(pTemp):
    fhTemp = pTemp * 1.8 + 32
    fhTemp = int(round(fhTemp,0))
    return fhTemp

def FtoC(pTemp):
    ctTemp = (pTemp - 32) / 1.8
    ctTemp = int(round(ctTemp,0))
    return ctTemp

# -----------------------------------------------------
# Write your code below this line
totalF = 0
totalC = 0
cTemp = 0
fTemp = 0

# update city temp
thisCity = input("Please enter City: ")
if thisCity in Tbl_city:
    thisIndex = Tbl_city.index(thisCity)
    thisTemp = input("Please enter temp in correct format: ")
    Tbl_temps[thisIndex] = thisTemp
else:
    print("City does not exist")
```

23 of 23

```
# display table header
layout = "{:16} {:^10} {:^10}"
print(layout.format("City", "Celsius", "Fahrenheit"))
print("-"*40)

cityCount = len(Tbl_city)

# take each city and temp
for i in range(cityCount):

    # get each scale and temp
    nextTemp = Tbl_temps[i]
    scale = nextTemp[0]
    theTemp = int(nextTemp[1:])

    # convert between temp scales
    if scale == "C":
        cTemp = theTemp
        totalC = totalC + cTemp
        fTemp = CtoF(theTemp)
        totalF = totalF + fTemp
    else:
        fTemp = theTemp
        totalF = totalF + fTemp
        cTemp = FtoC(theTemp)
        totalC = totalC + cTemp

    # display each row
    print(layout.format(Tbl_city[i], cTemp, fTemp))

# calculate averages
averageF = int(round(totalF / cityCount,0))
averageC = int(round(totalC / cityCount,0))

# display table footer
print("-"*40)
print(layout.format("Overall Average", averageC, averageF))
```

Type MS · The final 20-mark program

Answer 8 of 8 · Question 6 part whole · 20 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part whole in this mark scheme extract

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